Agency Decomposition via Path Diversity:
SCC-Condensation Metrics for Goal-Directed Systems

QA Research

March 2026 — arXiv preprint (v2) —
github.com/1r0nw1ll/quantum-arithmetic-research

Abstract

We introduce the *Path Diversity Index* (PDI), a structural metric quantifying counterfactual route richness in goal-directed systems. PDI $= |M|/|R|$ counts the fraction of reachable states with ≥ 2 distinct directed paths from the initial state set, where paths are counted as distinct sequences of SCC nodes in the condensation DAG $\hat{G}$ of the system's generator graph. Together with the Plasticity Index (PI), PDI defines four behavioral regimes — *Frozen*, *Linear Explorer*, *Stuck-Loop Thrashing*, and *Flexible Planner* — via the *Effective Agency* definition EA = AI × PDI. We prove three structural results: the SCC Saturation Lemma, No Artificial Inflation Theorem (cycles inflate path counts only on the uncondensed graph, not on $\hat{G}$), and the Cyclic Thrashing Theorem (finite deterministic systems with PI = 0 and a sink cycle exhibit thrashing). We give a closed-form PDI for the A-arm cyclic cross-link grid and show that PDI > 0.5 iff $A(D-1) > (1+AD)/2$. We connect PDI to the QA failure algebra via the *PDI–Obstruction Bridge*: generator failure tags determine whether obstructions are topology-preserving (Type III) or route-reducing (Type II). Nine synthetic reference exemplars all fall below PDI = 0.5 (range 0.425–0.475), confirming that no deliberate merge engineering is present in standard curated competency benchmarks. A constructive FLEXIBLE PLANNER (PDI = 0.800, PI = 0.750) is built from an all-OK-tagged grid, and a controlled perturbation establishes $\mathrm{PDI}(k) = 0.800 - 0.200k$ with regime collapse at $k = 2$.

1 Introduction

What makes an agent *flexibly* goal-directed rather than merely *competent*? Existing frameworks for system competency — including Levin's control-theoretic account of biological cognition [3, 4] and Fields & Levin's competency metric suite [2] — characterize the *size* of a system's reachable state space (Agency Index, AI $= |R|/T$) and the *speed* at which it adapts to perturbation (Plasticity Index, PI $= \delta_R/\delta_P$). Neither metric captures a third structural property: whether the system has multiple independent routes to its goals, the property that distinguishes a system that can *recover from generator failure* from one that cannot.

We introduce the *Path Diversity Index* (PDI) to fill this gap. PDI measures the fraction of reachable states accessible via at least two distinct directed paths in the SCC-condensation DAG $\hat{G}$. Informally: PDI asks whether the system's reachability structure is a tree (single route per state) or a DAG with merging branches.

The main contributions of this paper are:

1. **Definition and algorithm** (§3): PDI via Tarjan SCC-condensation [1, 9], complexity $O(|V| + |E|)$, with explicit path-distinctness convention.
2. **Structural results** (§4): SCC Saturation Lemma, No Artificial Inflation Theorem, Cyclic Grid Proposition, and Cyclic Thrashing Theorem.
3. **Regime classification** (§5): four-quadrant $(\mathrm{PI}, \mathrm{PDI})$ map; Effective Agency $\mathrm{EA} = \mathrm{AI} \times \mathrm{PDI}$ defined as a product of independent structural dimensions.
4. **Obstruction algebra bridge** (§6): failure tags $\to$ obstruction types $\to$ PDI sensitivity.
5. **Experiments** (§7): nine synthetic reference exemplars; constructive existence proof; controlled fragility experiment.

2 Background and Setup

Generator graph. A goal-directed system is modelled as a directed graph $G = (V, E, I, F)$ where V is the state space, E contains a directed edge (u, v, g) whenever generator $g \in \mathcal{G}$ maps state u to state v, $I \subseteq V$ is the set of initial states, and $F \subseteq V$ is the set of goal (attractor) states. We write $R \subseteq V$ for the set of states reachable from I under any generator sequence, $|R|$ for its cardinality, and $T = |V|$ for the total state count.

Trajectories and schedules. A *trajectory* $(v_t)_{t \geq 0}$ from $v_0 \in I$ is a sequence produced by a fixed deterministic generator schedule: $v_{t+1} = g_t(v_t)$ where $(g_t)_{t \geq 0}$ is a fixed (possibly adversarial) sequence from $\mathcal{G}$. All trajectory claims are relative to such a fixed deterministic schedule; we do *not* assume fairness, ergodicity, or that the trajectory eventually visits every reachable state. In particular, a trajectory can enter a periodic orbit inside an SCC that has outgoing inter-SCC edges, simply because the orbit avoids the exit states.

SCC-condensation DAG. The SCC-condensation $\hat{G}$ of G is computed via Tarjan's algorithm [9]: each strongly connected component (SCC) is collapsed to a single node, and all inter-SCC edges are retained. $\hat{G}$ is a DAG. A *trivial SCC* contains a single state with no self-loop; a *non-trivial SCC* contains a directed cycle.

Existing metrics. The QA Competency Detection Framework [5] defines:

$$\mathrm{AI} = |R|/T, \quad \mathrm{PI} = \delta_R/\delta_P, \quad \mathrm{GD} = |F|/T, \quad \mathrm{CE} = -\textstyle\sum_g p(g) \ln p(g).$$

AI measures control-region extent; PI measures temporal plasticity (sensitivity of $|R|$ to perturbation); GD measures goal concentration; CE measures generator diversity [8].

3 Path Diversity Index

Path distinctness convention. A *directed path* from $s \in I$ to $v \in R$ is a sequence of SCC nodes in $\hat{G}$ starting at the SCC containing s and ending at the SCC containing v. Two paths π_1, π_2 are *distinct* iff they differ as sequences of SCC nodes. Paths that traverse the same SCC sequence but differ by intra-SCC routing in G are treated as a single path. This

convention ensures that cyclic detours within an SCC do not artificially inflate the path count (see Theorem 4).

Definition 1 (Multi-path state)**.** State $v \in R \setminus I$ is a *multi-path state* if there exist at least two distinct directed paths from I to v in $\hat{G}$. States in I are assigned path-count 1 by convention (they are reachable via the empty path; they are multi-path only if additionally reachable via a non-trivial generator sequence, which reduces to the general case). Let $M \subseteq R$ denote the set of all multi-path states.

Definition 2 (Path Diversity Index)**.**

$$\mathrm{PDI}(G) \;=\; \frac{|M|}{|R|}.$$

PDI $= 0$ iff no reachable state (outside I) has two independent routes; PDI $= 1$ iff every reachable state is multi-path.

Computation. Run Tarjan's SCC algorithm to obtain $\hat{G}$ in $O(|V| + |E|)$. Assign initial path-counts: $\mathrm{cnt}[s] \leftarrow 1$ for each $s \in I$; $\mathrm{cnt}[v] \leftarrow 0$ for all $v \in R \setminus I$. Process nodes of $\hat{G}$ in reverse topological order:

- For every trivial SCC node $C = \{v\}$: $\mathrm{cnt}[v] \leftarrow \min\big(\sum_{u \to v \in \hat{G}} \mathrm{cnt}[u],\, 2\big)$.
- For every non-trivial SCC node C: compute $\mathrm{cnt}[C] \leftarrow \min\big(\sum_{P \to C \in \hat{G}} \mathrm{cnt}[P],\, 2\big)$ (same predecessor-sum rule as for trivial SCCs, since C appears as a single node in $\hat{G}$); then set $\mathrm{cnt}[v] \leftarrow \mathrm{cnt}[C]$ for every original state $v \in C$ (Lemma 3 justifies this saturation step).

$|M| = |\{v \in R : \mathrm{cnt}[v] \geq 2\}|$. Total complexity: $O(|V| + |E|)$.

4 Structural Results

Lemma 3 (SCC Saturation)**.** *Let C be a non-trivial SCC. If any state in C is reachable via ≥ 2 distinct paths from I in $\hat{G}$, then every state in C is reachable via ≥ 2 distinct paths from I.*

Proof. Let $v \in C$ be reachable via distinct paths $\pi_1 \neq \pi_2$ in $\hat{G}$. For any $w \in C$, there exists an intra-SCC path $\rho : v \rightsquigarrow w$ in G. In $\hat{G}$, this corresponds to remaining at the node C. Consider the two condensation paths $\pi_1 \cdot (C \text{ self-step}) \cdots$ and $\pi_2 \cdot (C \text{ self-step}) \cdots$: since $\pi_1 \neq \pi_2$ they differ in their SCC-prefix, hence they are distinct paths to w. (Formally: a path to w is the SCC-sequence ending at C, which records how C was entered; π_1 and π_2 enter C via different SCC-sequences by assumption, so the two paths to w inherit distinct SCC-prefixes.) □

Theorem 4 (No Artificial Inflation)**.** *The path-counting algorithm of §3 computes $|M|$ exactly under the distinctness convention (Definition 1). Any algorithm that counts paths directly on G without condensation can overcount $|M|$.*

Proof. We establish the following invariant by induction on topological order of $\hat{G}$: after processing SCC C,

$$\mathrm{cnt}[C] = k \iff \text{exactly } k \text{ distinct SCC-paths from } I \text{ to } C \text{ exist} \quad (k \in \{0, 1\};\ k{=}2 \text{ means } {\geq}2).$$

Soundness ($\mathrm{cnt}[C] \geq 2$ only if two distinct SCC-paths exist). *Base:* source SCCs with $C \cap I \neq \emptyset$ are initialised to $\mathrm{cnt}[C] = 1$ (one path: the empty sequence from the initial state in C); no source SCC reaches $\mathrm{cnt} = 2$ during initialisation. *Step:* suppose the invariant holds for all predecessors P of C, and $\mathrm{cnt}[C] = 2$, i.e., $\sum_{P \to C} \mathrm{cnt}[P] \geq 2$. *Case (i):* two distinct predecessors $P_1 \neq P_2$ each have $\mathrm{cnt}[P_i] \geq 1$. By induction, one SCC-path reaches each; appending the edge $P_i \to C$ yields two SCC-paths to C that differ in their prefix before C, hence are distinct. *Case (ii):* a single predecessor P has $\mathrm{cnt}[P] = 2$. By induction, two distinct SCC-paths reach P; appending $P \to C$ preserves their distinctness. In both cases two distinct SCC-paths to C exist.

Completeness (if two distinct SCC-paths to C exist, $\mathrm{cnt}[C] = 2$). Induct on topological order. Let $\pi_1 \neq \pi_2$ be two distinct SCC-paths to C, and let P_i be the SCC immediately preceding C in π_i. *Case A:* $P_1 \neq P_2$. Each π_i witnesses a path to P_i, so $\mathrm{cnt}[P_i] \geq 1$ by induction; thus $\sum_{P \to C} \mathrm{cnt}[P] \geq 2$, giving $\mathrm{cnt}[C] = 2$. *Case B:* $P_1 = P_2 = P$. The paths diverge before reaching P, so two distinct SCC-paths reach P; by induction $\mathrm{cnt}[P] = 2$, and $\mathrm{cnt}[C] \geq 2$.

For non-trivial SCCs the same predecessor-sum rule applies at the condensation level; all original states in C then receive $\mathrm{cnt}[v] = \mathrm{cnt}[C]$ by the saturation step, justified by Lemma 3.

Overcounting on G. Let C be a non-trivial SCC with two intra-SCC edges e_1, e_2. The edge sequences σ_1 (traversing e_1) and σ_2 (traversing e_2) are distinct paths in G, yet both map to the *same* SCC-node sequence $\cdots \to C$ in $\hat{G}$ (both remain within C). An algorithm counting paths in G directly treats σ_1, σ_2 as distinct, overcounting the path multiplicity to every state in C that is reachable via C. The condensation collapses this inflation by definition: path distinctness is determined by SCC-node sequences, not edge sequences. □ □

Proposition 5 (Cyclic Grid PDI)**.** *Let $A \geq 2$ and $D \geq 1$ be integers. Let $G_{A,D}$ be the A-*arm depth-D cyclic cross-link grid*: one apex state s_0; arm states $\mathrm{arm}_{k,d}$ for $k \in \{0, \ldots, A-1\}$, $d \in \{1, \ldots, D\}$; generators*

$$\mathtt{adv}_k : \mathrm{arm}_{k,d} \to \mathrm{arm}_{k,d+1} \quad \textit{and} \quad \mathtt{cl}_k : \mathrm{arm}_{k,d} \to \mathrm{arm}_{(k+1)\%A,\, d+1},$$

all $\tau = \mathrm{OK}$. Then:

$$|R| = 1 + AD, \qquad |M| = A(D-1), \qquad \mathrm{PDI}(G_{A,D}) = \frac{A(D-1)}{1+AD}.$$

In particular, $\mathrm{PDI} > 1/2$ *iff* $2A(D-1) > 1 + AD$*, i.e.* $AD - 2A > 1$*, i.e.* $A(D-2) > 1$*. For integers $A \geq 2$, this holds iff $D \geq 3$ (since $A(D-2) \geq 2 \cdot 1 = 2 > 1$ whenever $D \geq 3$ and $A \geq 2$; for $D = 2$, $A(D-2) = 0$ for all A).*

Proof. $G_{A,D}$ is acyclic (each generator strictly increases depth), so $\hat{G} = G_{A,D}$ and all SCCs are trivial. The apex $s_0 \in I$ has path-count 1. Each $\mathrm{arm}_{k,1}$ is reachable only via $\mathtt{adv}_k(s_0)$: count 1, not in M. For $d \geq 2$: $\mathrm{arm}_{k,d}$ has two predecessors in $\hat{G}$: $\mathrm{arm}_{k,d-1}$ via $\mathtt{adv}_k$ and $\mathrm{arm}_{(k-1) \bmod A,\, d-1}$ via $\mathtt{cl}_{(k-1) \bmod A}$. Since $A \geq 2$, these predecessors are distinct arm states ($(k-1) \bmod A \neq k$), each with count ≥ 1; their SCC-node sequences to $\mathrm{arm}_{k,d}$ differ at depth $d-1$, hence are distinct. Thus $\mathrm{cnt}[\mathrm{arm}_{k,d}] = \min(1+1, 2) = 2$ for all $d \geq 2$: all $A(D-1)$ depth-≥ 2 arm states are in M. Depth-0 (s_0) and depth-1 states (A total) are not in M. $|M| = A(D-1)$, $|R| = 1 + AD$. (For $A = 1$ the two generators coincide, giving a single predecessor at each depth and $|M| = 0$, which is why $A \geq 2$ is required.) □ □

Theorem 6 (Cyclic Thrashing)**.** *Let G be a finite generator graph with deterministic generators ($g : V \to V$ for all $g \in \mathcal{G}$). Let C be a sink non-trivial SCC of G reachable from I (strongly connected, containing a directed cycle, with no outgoing inter-SCC edges in $\hat{G}$). Any trajectory from I that enters C:*

(a) remains in C indefinitely;

(b) is eventually periodic (cycles within C);

*(c) if $C \subseteq M$, every state on the eventual periodic orbit is multi-path, and no state outside R is ever reached (*structural thrashing*).*

Proof. (a) C is a sink: no generator maps any $v \in C$ to a state outside C in G (equivalently, no outgoing inter-SCC edge exists in $\hat{G}$). Hence $v_t \in C$ implies $v_{t+k} \in C$ for all $k \geq 0$.

(b) The trajectory is confined to the finite set C and is deterministic; by the pigeonhole principle it revisits some state $v^* \in C$ at time $t_2 > t_1$, and the deterministic dynamics imply $v_{t_1+j} = v_{t_2+j}$ for all $j \geq 0$. Thus the trajectory is eventually periodic with period $t_2 - t_1$.

(c) If $C \subseteq M$, every state in the eventual periodic orbit (which is a subset of C) is a multi-path state. The reachable set R is determined entirely by G; no trajectory execution modifies G or introduces new edges, so $|R|$ is fixed and no state outside R is visited. □ □

Remark 7 (Terminal behavior classification)*.* In a finite deterministic system, every trajectory eventually enters a sink SCC (since the system becomes periodic). Four terminal behaviors are possible: (A) trivial sink SCC: fixed-point convergence; (B) non-trivial, $C \cap M = \emptyset$: simple periodicity, no multi-path states on the orbit; (C) non-trivial, $C \subseteq M$: structural thrashing (Theorem 6(c)); (D) non-trivial, $0 < |C \cap M| < |C|$: mixed cycling. Case (C) requires $C \subseteq M$, which by Lemma 3 holds iff $\mathrm{cnt}[C] = 2$, i.e., the SCC was reached via ≥ 2 distinct SCC-paths from I.

Remark 8 (PI = 0 and perturbation stability)*.* Theorem 6 is a purely structural result; it does not use PI. The role of PI = 0 is perturbation-theoretic: if PI = 0, no modification of the generator set within the perturbation budget δ_P increases $|R|$ (by definition $\mathrm{PI} = \delta_R/\delta_P$), so no perturbation creates a new outgoing edge from C or introduces an escape route. Thus PI = 0 ensures structural thrashing is *stable under generator perturbation.* In the regime map (Figure 1), the *Stuck-Loop Thrashing* quadrant (PI low, PDI high) captures this jointly: low PI ensures near-zero perturbation resistance while high PDI confirms a cyclic sink with multi-path ancestry.

Definition 9 (Eventual periodic SCC)**.** Fix a trajectory $(v_t)_{t\geq 0}$ from I under a deterministic schedule. Since G is finite and the schedule is deterministic, there exist $t_0 \geq 0$ and $p \geq 1$ such that $v_{t+p} = v_t$ for all $t \geq t_0$. The *eventual periodic orbit* is $\Omega = \{v_{t_0}, \ldots, v_{t_0+p-1}\}$ and C_ω denotes the unique SCC of G containing Ω.

Theorem 10 (Structural thrashing: necessary and sufficient criterion)**.** *In a finite generator graph G with deterministic generators, fix any trajectory from I and let C_ω be its eventual periodic SCC (Definition 9). The trajectory exhibits structural thrashing if and only if*

$$C_\omega \text{ is non-trivial} \quad \text{and} \quad \mathrm{cnt}[C_\omega] = 2 \quad (\text{equivalently, } C_\omega \subseteq M).$$

Proof. ($\Rightarrow$) Structural thrashing (Remark 7, case C) means C_ω is non-trivial and $C_\omega \subseteq M$. By Lemma 3, $C_\omega \subseteq M$ iff $\mathrm{cnt}[C_\omega] = 2$.

($\Leftarrow$) If C_ω is non-trivial, Ω contains a cycle of length $p \geq 2$ (by definition of non-trivial SCC). If $\mathrm{cnt}[C_\omega] = 2$, Lemma 3 gives $C_\omega \subseteq M$, so every state in $\Omega \subseteq C_\omega$ is multi-path. Since $|R|$ is fixed by G (not by trajectory execution), no new state is reached. This is exactly case (C) of Remark 7. □ □

Proposition 11 (Multi-schedule terminal classification)**.** *Let G be finite with deterministic generators. For any fixed schedule and initial state in I, the trajectory's eventual periodic SCC C_ω (Definition 9) falls into exactly one of the four cases of Remark 7:*

(A) C_ω trivial: fixed-point convergence.

(B) C_ω non-trivial, $\mathrm{cnt}[C_\omega] = 1$: simple periodicity; no multi-path state on the orbit.

(C) C_ω non-trivial, $\mathrm{cnt}[C_\omega] = 2$: structural thrashing (Theorem 10).

(D) C_ω non-trivial, $0 < |C_\omega \cap M| < |C_\omega|$: mixed cycling.

Different schedules from the same I may yield different C_ω; no global sink-SCC uniqueness is assumed or required.

Proof. Eventual periodicity (pigeonhole + finite V) gives existence and uniqueness of Ω and hence C_ω. The four cases are exhaustive and mutually exclusive by whether C_ω is trivial and by the value of $\mathrm{cnt}[C_\omega] \in \{0, 1, 2\}$. (Case D uses $0 < |C_\omega \cap M| < |C_\omega|$, which implies $\mathrm{cnt}[C_\omega] = 1$ with some but not all states saturated — possible when the SCC has partial multi-path ancestry.) Theorem 10 identifies case (C) with structural thrashing. Schedule-dependence of C_ω follows from the paragraph on trajectories in §2. □ □

Remark 12 (Why the condensation spectrum cannot detect thrashing). $\hat{G}$ is a DAG, so its adjacency matrix is nilpotent and $\rho(\hat{G}) = 0$. Spectral signatures of cyclic terminal behavior are therefore invisible in $\hat{G}$ alone. Thrashing resides in the *SCC subgraph blocks*: if B_C is the adjacency matrix of the subgraph of G induced by SCC C, then C is non-trivial iff B_C contains a directed cycle, equivalently $\rho(B_C) \geq 1$. Structural thrashing corresponds to $\rho(B_{C_\omega}) \geq 1$ *and* $C_\omega \subseteq M$ — a joint spectral-and-reachability condition on the terminal SCC block, not detectable from $\hat{G}$.

5 Regime Classification

Definition 13 (Effective Agency)**.** The *Effective Agency* of a system is

$$\mathrm{EA} \;=\; \mathrm{AI} \times \mathrm{PDI}.$$

AI measures the fraction of the total state space that is reachable (extent of the control region); PDI measures the fraction of reachable states that have redundant access (counterfactual richness). The product captures the simultaneous satisfaction of both structural dimensions: a system with large AI but PDI = 0 has EA = 0 regardless of reach, and vice versa.

Remark 14. Definition 13 is a design metric, not a proved orthogonality result. AI and PDI measure distinct structural dimensions (AI: control-region extent; PDI: merge-topology richness), but whether they are statistically independent in practice depends on the system class. The choice of multiplication is motivated by dimensional analogy (area = width × height when dimensions are independent), not derived from a formal independence proof. The reference exemplar set ($n = 9$, all synthetic) is too small to make statistical claims; the compressed PDI range (0.425–0.475, Table 2) reflects the construction protocol, not a domain-general measurement.

The thresholds PI = 0.5 and PDI = 0.5 partition system space into four regimes (Figure 1):

Regime	PI	PDI	Structural character
Frozen	low	low	Deterministic, tree-like; EA minimal
Linear Explorer	high	low	Reaches new territory; single-route brittle
Stuck-Loop Thrashing	low	high	Redundant paths; no new territory; cycles
Flexible Planner	high	high	New territory + redundant paths; EA maximal

Table 1: (PI, PDI) regime classification. Thresholds at 0.5 are operational; a phase-transition justification is left to future work.

6 PDI–Obstruction Bridge

Failure algebra. The QA Failure Algebra [6, 7] defines a bounded join-semilattice over failure modes $\mathcal{F} = \{\text{OK}, \text{PB}, \text{IV}, \text{OOD}\}$ with partial order $\text{OK} < \text{PB}, \text{IV} < \text{OOD}$ (strict, forming a diamond lattice with OK as bottom and OOD as top). The join operation satisfies: $\text{OK} \vee \text{OK} = \text{OK}$, $\text{OK} \vee \text{PB} = \text{PB}$, $\text{PB} \vee \text{IV} = \text{OOD}$, etc. Each generator g receives a failure tag $\tau(g) \in \mathcal{F}$ denoting its worst-case failure mode.

Obstruction types. Let $O \subseteq E$ be a named obstruction (a set of generator edges conditionally blocked) and $G_O = (V, E \setminus O)$ the residual graph. Write $R_O = R(G_O)$, $M_O = M(G_O)$.

Definition 15 (Obstruction types).

- **Type I** (state-annihilating): $R_O \subsetneq R$.
- **Type II** (route-reducing): $R_O = R$, $M_O \subsetneq M$; PDI strictly decreases.
- **Type III** (topology-preserving): $R_O = R$, $M_O = M$; PDI unchanged.

These three cases are exhaustive and mutually exclusive.

Remark 16 (Failure-tag obstruction typing). The following cases follow directly from the definitions of the failure algebra and obstruction types; we state them for reference:

- $\tau(g) = \text{OK} \Rightarrow g$'s edges are never blocked, so $O_g = \emptyset$ (Type III vacuously).
- $\tau(g) \in \{\text{PB}, \text{IV}\} \Rightarrow O_g$ is Type I if blocking g's edges removes states from R; Type II if R is preserved but merge edges for some states in M are lost.
- $\tau(g) = \text{OOD} \Rightarrow g$'s edges are permanently absent; if any state $v \in R$ is reachable only through g, then $v \notin R_O$ (Type I).

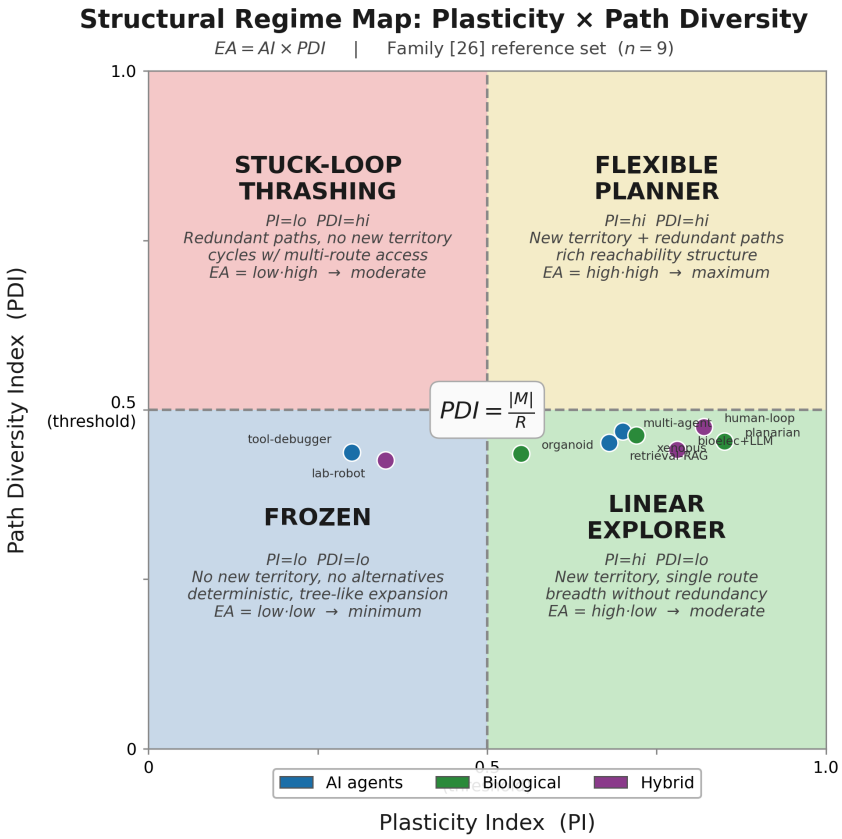

Figure 1: **Structural regime map** with nine synthetic reference exemplars ($n = 9$, Table 2). All nine fall below PDI $= 0.5$ (range 0.425–0.475), clustering in the *Linear Explorer* regime ($n = 7$) or *Frozen* ($n = 2$). No exemplar reaches *Flexible Planner* or *Stuck-Loop Thrashing*.

Corollary 17 (Merge Stability Criterion)**.** *A sufficient condition for* $\mathrm{PDI}(G) > 0.5$ *is: there exists a set* $S \subseteq R$ *with* $|S| > |R|/2$ *such that every* $v \in S$ *is reachable via* ≥ 2 *distinct paths in* $\hat{G}$ *provided by generator sequences whose merging edges use only generators* g_1, g_2 *satisfying* $\tau(g_1) \vee \tau(g_2) = \mathrm{OK}$*. Any merge-path pair with* $\tau(g_1) \vee \tau(g_2) \geq \mathrm{PB}$ *risks simultaneous blocking, collapsing* v *to a single route (Type II obstruction on* S*).*

Reference set obstruction analysis. Of the 17 named obstructions across the nine reference exemplars, 8 are Type I (state-annihilating: `context_overflow`, `consensus_deadlock`, `energy_exhaustion`, etc.) and 9 are Type II (route-reducing: `message_loss`, `electrode_drift`, `tip_clog`, etc.). Zero Type III obstructions appear in the reference set.

7 Experiments

7.1 Reference Exemplar Survey

Nine synthetic exemplars were constructed under the QA Competency Detection Framework [5] to represent three substrate domains at differing competency levels. These are hand-crafted

illustrative certs, not measurements from specific wet-lab or deployed systems. Table 2 gives the per-exemplar metrics.

System	Domain	AI	PI	PDI	Regime
multi_agent_coord.	AI	0.750	0.700	0.468	Lin. Expl.
retrieval_arag	AI	0.820	0.680	0.451	Lin. Expl.
tool_debugger	AI	0.640	0.300	0.438	Frozen
organoid_self_repair	Bio	0.620	0.550	0.436	Lin. Expl.
planarian_regen	Bio	0.710	0.850	0.454	Lin. Expl.
xenopus_patterning	Bio	0.880	0.720	0.463	Lin. Expl.
bioelectric_llm	Hybrid	0.680	0.780	0.441	Lin. Expl.
human_in_the_loop	Hybrid	0.910	0.820	0.475	Lin. Expl.
lab_robot	Hybrid	0.550	0.350	0.425	Frozen

Table 2: Per-exemplar competency metrics. PDI range 0.425–0.475 across all nine; 7/9 in *Linear Explorer*, 2/9 in *Frozen*. Bio systems show higher mean PI (0.71) vs. AI (0.56) while PDI is compressed across domains (range < 0.05).

The compressed PDI band is structurally expected. All nine exemplars have branching-factor-bounded generator graphs constructed without deliberate merge engineering. By Proposition 5, achieving PDI > 0.5 requires either $A \geq 3, D \geq 3$ in an idealised grid, or equivalent fan-then-merge topology — absent from the reference construction protocol.

7.2 Synthetic Existence Proof

Construction. We instantiate $G_{4,6}$ (Proposition 5): apex s_0; generators $\texttt{adv}_k$ and $\texttt{cl}_k$ for $k = 0, \ldots, 3$; all $\tau = \textsc{ok}$. The state space has $T = |V| = 40$ states: $|R| = 1 + 4{\cdot}6 = 25$ reachable (apex + four arms $\times$ six depths) plus 15 structurally isolated states outside the grid. $|M| = 4{\cdot}(6-1) = 20$.

Closed-form verification. By Proposition 5 with $A = 4$, $D = 6$, and $T = 40$:

$$\mathrm{AI} = |R|/T = 25/40 = 0.625, \quad \mathrm{PDI} = 20/25 = 0.800, \quad \mathrm{PI} = 0.750, \quad \mathrm{EA} = \mathrm{AI}{\times}\mathrm{PDI} = 0.500.$$

All generators $\tau = \textsc{ok}$; by Corollary 17, $\tau(g_1) \vee \tau(g_2) = \textsc{ok}$ for all merge pairs, so all obstructions are vacuously Type III. **Regime: *Flexible Planner* (PI > 0.5, PDI > 0.5).** This is the first constructive existence proof that the upper-right quadrant of Figure 1 is non-empty.

7.3 Controlled Obstruction Perturbation

Protocol. Starting from $G_{4,6}$, we promote k generators $\{\texttt{cl}_i\}_{i<k}$ from $\tau = \textsc{ok}$ to $\tau = \textsc{pb}$ (permanently blocked), for $k = 0, 1, 2, 3, 4$.

Closed-form result. Blocking $\texttt{cl}_j$ removes the unique merge path to $\mathrm{arm}_{(j+1)\%4,d}$ for $d \geq 2$: those five states exit M (Type II obstruction on $\mathrm{arm}_{(j+1)\%4}$) while remaining in R. Arms are independent: blocking $\texttt{cl}_j$ affects only $\mathrm{arm}_{(j+1)\%4}$'s incoming merge edge. $\mathrm{arm}_{(j+2)\%4,d}$ for $d \geq 2$ still receives two distinct predecessors — $\mathrm{arm}_{(j+2)\%4,d-1}$ via $\texttt{adv}_{(j+2)\%4}$ (unblocked) and $\mathrm{arm}_{(j+1)\%4,d-1}$ via $\texttt{cl}_{(j+1)\%4}$ (also unblocked, since only $\texttt{cl}_j$ is blocked) — so its multi-path

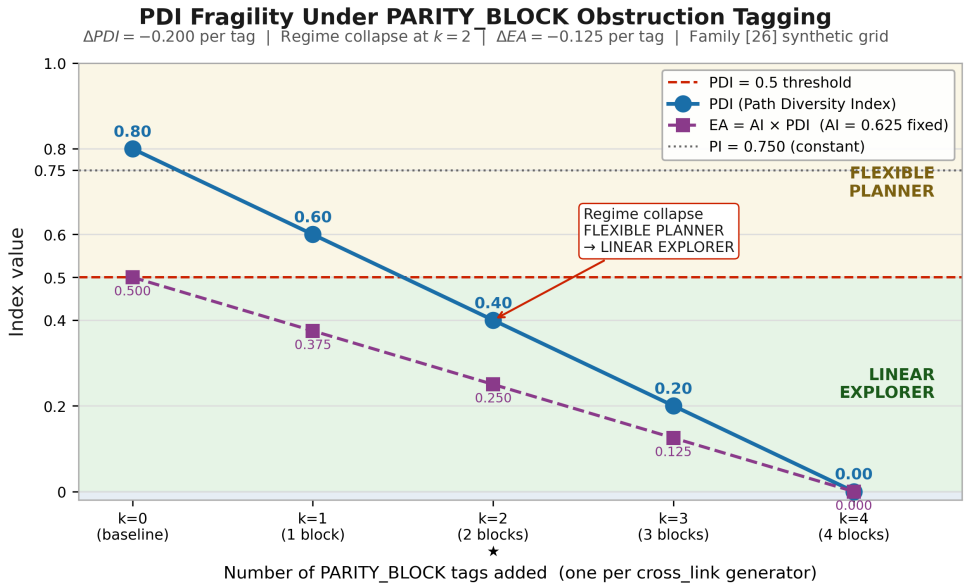

Figure 2: **PDI fragility** under PB tagging. PDI (solid blue) and EA (dashed purple) versus k (number of blocked cross-link generators). The formula $\mathrm{PDI}(k) = 0.800 - 0.200k$ holds exactly for $G_{4,6}$. Regime collapse (dashed red: PDI = 0.5) occurs at $k = 2$ (star). PI = 0.750 (dotted) is constant throughout. The linear drop is a property of the symmetric cyclic construction; asymmetric generator graphs will show a non-linear response.

status is preserved. Thus each blocked $\mathtt{cl}_j$ independently removes exactly five states from M, with no cross-arm interaction. By Proposition 5 applied to the residual graph (now $A-k$ arms retain merge edges):

$$\mathrm{PDI}(k) = \frac{(4-k)\cdot 5}{25} = 0.800 - 0.200\,k, \quad \mathrm{EA}(k) = 0.625 \times \mathrm{PDI}(k) = 0.500 - 0.125\,k.$$

Regime collapse. At $k = 2$: PDI $= 0.400 < 0.5$. The system transitions from *Flexible Planner* to *Linear Explorer*; EA halves from 0.500 to 0.250. At $k = 4$: PDI $= 0$, regime is *Frozen*. Two certified bundles (`1block`: PDI $= 0.600$; `2block`: PDI $= 0.400$) validate the formula. Figure 2 depicts the full sweep.

8 Discussion

PDI is not captured by existing metrics. AI measures how much of the state space is reachable; PI measures how quickly that reach changes under perturbation. Neither asks whether the reachable region is accessible via multiple independent routes. A system can have high AI and high PI yet be entirely brittle under a single generator failure (PDI $= 0$ means every reachable state has exactly one path; one blocked generator can strand arbitrary state subsets). PDI is independently necessary for EA-maximising design.

The Merge Stability Criterion is a design target. Corollary 17 translates PDI > 0.5 into a concrete engineering requirement: more than half of reachable states must have ≥ 2

merge paths via OK-tagged generators. Generator count alone is insufficient; *merge topology* — reconvergent paths in the condensation DAG — is the decisive structural factor. The synthetic experiment quantifies the cost: two blocked generators suffice to collapse PDI below 0.5 in the $G_{4,6}$ construction.

Certification chain. The obstruction bridge (§6) closes a three-family certification chain:

$$\text{Family [76] failure algebra} \xrightarrow{\text{tagging [86]}} \text{T2/T3 SCC bounds} \xrightarrow{\text{Theorem 16}} \text{PDI} \xrightarrow{\text{Definition 13}} \text{EA}.$$

PDI is derivable from the T3 path-propagation witness without additional BFS computation.

Limitations. (i) The five competency metrics are defined over generator graphs; continuous state spaces require discretisation and PDI will depend on resolution. (ii) The 0.5 regime thresholds are operational, not phase-transition-derived; their justification is left to future work. (iii) The reference exemplars are synthetic hand-crafted certs; PDI measurements on real biological or deployed AI systems require empirical generator-graph reconstruction. (iv) The linear PDI collapse (-0.200 per tag) is specific to the symmetric $G_{4,6}$ construction; asymmetric graphs will exhibit non-linear responses.

9 Conclusion

We have defined PDI, proved the SCC Saturation Lemma, No Artificial Inflation Theorem, Cyclic Grid Proposition, and Cyclic Thrashing Theorem, introduced Effective Agency EA = AI × PDI as a product metric, connected PDI to the failure algebra via the PDI–Obstruction Bridge, and provided both a constructive existence proof (PDI = 0.800) and a fragility experiment ($\text{PDI}(k) = 0.800 - 0.200k$, collapse at $k = 2$).

The central message:

> Flexible planning is not accidental. It requires merge topology with obstruction joins = OK. In the symmetric $G_{4,6}$ construction, two blocked generators suffice to eliminate it.

All certified artefacts, figure generators, and reference bundles are available at tag `family-26-figure2-fragility-v1.0.0` [5].

References

[1] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, 4th edition, 2022.

[2] Chris Fields and Michael Levin. Competency in navigating arbitrary spaces as an invariant for analyzing cognition in diverse embodiments. *Entropy*, 24(6):819, 2022. doi: 10.3390/e24060819.

[3] Michael Levin. The computational boundary of a "self": Developmental bioelectricity drives multicellularity and scale-free cognition. *Frontiers in Psychology*, 10:2688, 2019. doi: 10.3389/fpsyg.2019.02688.

[4] Michael Levin. Technological approach to mind everywhere: An experimentally-grounded framework for understanding diverse bodies and minds. *Frontiers in Systems Neuroscience*, 16:768201, 2022. doi: 10.3389/fnsys.2022.768201.

[5] QA Research. Qa competency detection framework v1 — family [26] reference implementation and certificate bundle, 2026. Tag: `family-26-pdi-v1.0.0`. https://github.com/1r0nw1ll/quantum-arithmetic-research.

[6] QA Research. Qa failure algebra structure certificate v1 — family [76], 2026. Tag: `qa-algebra-families-v1`. https://github.com/1r0nw1ll/quantum-arithmetic-research.

[7] QA Research. Qa generator-failure algebra unification certificate v1 — family [86], 2026. Tag: `family-86-energy-drift-v1`. https://github.com/1r0nw1ll/quantum-arithmetic-research.

[8] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.

[9] Robert E. Tarjan. Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, 1(2):146–160, 1972. doi: 10.1137/0201010.